

Group Problem Set 19: Systems of Particles Center of Mass and Momentum

We use center of mass of a body to represent the location around which the object obeys newtonian physics. It is much easier to choose a central location on an object to describe its motion than to write equations of motion for each of its parts. The center of mass of a two body system can be found by taking the weighted sum of the individual mass locations and dividing by the object's total mass:

$$x_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

We can extend the center of mass equation to a more generalized form. When we do this, we aren't limited by the number of objects we consider part of the system:

$$x_{CM} = \frac{\sum_{j=1}^n m_j x_j}{\sum_{j=1}^n m_j} = \frac{1}{M_{tot}} \sum_{j=1}^n m_j x_j$$

Marshmallows and skewers are provided to assist in problem set up and calculations.

1. Jim wants to 'pump it up' and takes up bench pressing weights at the gym. He is very confident of his abilities and places 100 kg weights at each end of a 1 m barbell. So that Jim doesn't hurt himself, where should Jim lift the barbells. Use the marshmallows and skewers provided to verify your predictions?
2. Jim's friend Karen isn't convinced that Jim has properly placed the weights. Karen redistributes the weights so that 125 kg are attached to one end and 75 kg are attached to the opposite end of the barbell. Draw this setup, including distances and apply the equation for x_{CM} to calculate the center of mass for the barbell system Karen has set up. A system is equally balanced in all directions at its center of mass. If Jim were to pick the bar up at its center, would he be able to easily balance the bar?
3. Jim's barbell can't hold any more weight at its ends but something could be added along the barbell. Hypothesize where Jim might add weight to Karen's setup so that he can again easily lift the bar at its center. Predict a location and mass that would help redistribute the weight so that the center of mass is at the center of the bar.

Now let's revisit Newton's Second Law:

$$\vec{F}_{total} = \sum \vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt}$$

Because we are now thinking of things in terms of masses at locations, we can incorporate this new idea into the sum of forces:

$$\vec{F}_{total} = \sum \vec{F} = m\vec{a} = \frac{d(m\vec{v})}{dt}$$

We've created a new formulation for how an object's motion is tied to its position, $m\vec{v}$. This new relation is called momentum and is represented in vector form:

$$momentum = m\vec{v} = \vec{p}$$

From here, it is easy to relate momentum of an object to the sum of forces.

$$\vec{F} = \frac{d\vec{p}}{dt}$$

4. Extrapolate from the above relations to answer the questions below:

- (a) In your own words, define momentum.
- (b) Write the relationship for momentum using the correct vector notation.
- (c) Write the relationship for force with respect to momentum.
- (d) What is required to happen to momentum of a system for the sum of forces to be non-zero?
- (e) If there is no change in momentum and mass is constant, what can be said about the associated force?
- (f) Is the reference frame of an object with a constant momentum inertial or non-inertial?
- (g) Rockets burn fuel to propel forward. Force is required to maintain a forward motion and without it, the rocket would be subjected to projectile motion under the force of gravity and would fall back to the earth. A rocket is shot and continues to travel at a constant velocity away from the earth. Reevaluate the relationship between force and momentum and discuss what permits the rocket to generate propulsion.

5. The center of mass of a system does not require that the particles in the system are attached. For this exercise, you are going to calculate the center of mass of your team.
- (a) Using the grid below, estimate and plot the locations of your group members. Represent each one with a specific shape and write the coordinates to the side.
 - (b) Next estimate the mass of each member in your team (when in doubt, you can always use Homer Simpson's mass... 100 kg). These masses don't need to be super correct but should all be different and keep it under 500 kg per person.
 - (c) Now calculate the center of mass of your team. The center of mass in 2 dimensions should have an x location and y location. This requires you to calculate the center of mass with respect to x and then again with respect to y. They will not be the same.
 - (d) Finally, add the center of mass to your diagram.

